ROYAL CANADIAN NAVY

High Voltage Safe System of Work (SSOW) Competent Person

QUALIFICATION STANDARD AND PLAN (QSP)

Marine Technician (MAR TECH) (00379)
Weapons Engineering Technician (WENG TECH) (00366)
Boatswain (BOSN) (00105)

Common Specialty Qualification - (AMBW)

MITE Course IDs:
Competent Person (CP) – DL 123124 & OJPR 123125

RECORD OF AMENDMENTS

This QSP is developed under the authority of the Royal Canadian Navy (RCN) through the Training Authority (TA), Naval Personnel and Training Group (NPTG) and the Commanding Officer of the Naval Training Development Centre (NTDC). The approval or revision date should appear in the footer of every page as an additional means of verification. Details on the change(s) must be amplified in Annex J “Course History”.

Revision	Approval Date	Authority
1.0 (QSPWB March 2021)	04 July 2023	CNTG

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FOREWORD

1 This training document is developed under the authority of the Commander Naval Personnel and Training Group (CNPTG) and in accordance with the following references:

a. A-P9 - 050 Series, Canadian Forces Manual of Individual Training and Education (CFITES);

b. A-PD-055-002/PP-003 Canadian Forces Officer General Specification;

c. A-PD-055-002/PP-002 Non-Commissioned Member General Specification;

d. A-PD-055-002/PP-001 Name of Occupation Job Based Specification or Job Based Occupational Specification;

e. A-PD-055-003/PQ-001 Name of Specialty Specification; and

f. NAVORD 4500-0 Structured Delegation; and

g. NPTORD 4500-22 Qualification Standard and Plan Production and Management.

2 This QSP consists of specific Performance Objectives (POs), Activity and Assessment Plan (AAP) in Chapter 3. A trainee will achieve the qualification when the standards listed in the POs have been met.

3 This QSP is effective upon approval and supersedes previous versions.

Note: HV AOPV training is a new capability and this is the first QSP version.

4 Suggestions for amendments to this QSP may be initiated by anyone and shall be forwarded in writing to the CCA (NFS(P) DDE Div Comd), who will forward to NPTG for approval, as appropriate.

5 The designated TEs for this training at this time are not applicable the HV interim training consists of eLearning via distributed learning and OJPR on ship.

TE	Phase / Module	MITE Course IDs	NQual
NFS(A/P)	DL	123124	
NFS(A/P)	OJPR	123125	AMBW

CHAPTER 1 TRAINING MANAGEMENT DETAILS

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY COMMANDER NAVAL PERSONNEL AND TRAINING GROUP (CNPTG)

MARPAC RDIMS # 1381596
Chap 1 approved 04 07 2023
Chap 2, 3 approved 04 07 2023

AIM

1 The aim of this Qualification Standard and Plan (QSP) is to establish the requirements and training for the **Competent Person (CP)**. This QSP is designed to assist the training system in developing, conducting, evaluating, and validating the training.

Competent Person (CP): CP qualified persons have sufficient technical knowledge to enable the person to avoid danger from electrical hazards. CP qualified persons may enter HV compartments and may act as a Person in Charge to work on dead HV equipment. All members of the Naval Technical Department shall hold this qualification at minimum. Other personnel where appropriate may also hold this qualification.

Interim State Training Overview:

As part of the interim state training, the HV Individual Training & Education (IT&E) will be delivered through various delivery models based on HV qualification:

- CP: OJPR on HMC Ship (AOPV)

COURSE DURATION

2 Course duration is calculated in accordance with NPTORD 4500-22 Annex B. Detailed duration with resource allocation is found in Annex G of this QSP. Note, the DL timings are approximations and will need to be validated during post pilot course review.

Allocation of ATT	**Minutes**
Lesson Objectives (LOs) / Teaching Points Competent Person	CP: 345 mins
Performance Checks (PCs) Competent Person	35 mins per trainee
ATT Sub-total	
Allocation of BAT, TAT, TTT, PT (Course > 31 days)	
Base Administrative Time (BAT)	N/A
Training Administrative Time (TAT)	N/A
Training Travel Time (TTT)	N/A
Physical Fitness (PT)	N/A
Sub-total	N/A
Course Duration Calculation (ATT+BAT+TTT+TAT+PT) **Total mins/400 = # days (rounded up)**	
Total Training Days	N/A

RESOURCES

3 The following table details key personnel required to conduct the training:

Required Course Staffing				
Position	Rank	Qualifications Required	Qty	PO / EO
OJPR Trainer	PO1	Sr. Electrical Propulsion Mgr AE and AP Qualifications	1	PO 001 & PO 002
OJPR Trainer	Lt(N)	Naval Technical Officer AE and AP Qualifications	1	PO 003

4 The following table details the critical training resources to conduct the training. It should be read in conjunction with the recommended course timetable in Annex G of this QSP.

Critical Training Resources	PO/EO
1. DND computer tablets (only if trainee does not have a personal computer at home to conduct DL) 2. AOPV ship 3. Virtual Task trainer 4. OJPR Package	PO 001

COURSE CAPACITY

5 Planning for the conduct of training is based on the guidelines in CFITES Vol 5 Annex D (OJPR coach to Learner Ratios). Justification can be found in Annex C.

Distributed Learning for AP members

Note: given the eLearning courses are not facilitated and conducted via distributed learning, there are no requirements for max, min, and optimum.

OJPR on AOPV

	CP
Max	6
Min.	1
Optimum	4

Notes: OJPRs shall only be conducted after the successful completion of eLearning for EO 001.01 and EC 001.01.

MITE

6 Loading. The authority to load members onto this course lies with the NFS. NPTG is to be included as an INFO addressee on all loading messages.

7 Scheduling. Each session of this course shall be scheduled IAW NPTORD 4500-1 - Individual Training and Education Policy and NPTORD 4500-40 Quantity Control Management for the Naval Training System (NTS). It is to be scheduled in MITE by the NFS conducting the training and NPTG through the Quantity Control section. The course length in MITE is to be based on the course duration, planned annual leave, training breaks, weekends and training system holidays. Course sessions for Naval Reserve Force may be scheduled during training system holidays.

8 Prerequisites. The loading authority must ensure that trainees have the following prerequisites before being selected for this course:

Distributed Learning:

a. CP eLearning Course; and
b. CP OJPR package.

9 Qualification. Successful completion of this course will grant the trainee the national CP **High Voltage Training** (CPHVQ) qualification.

COURSE REPORTING

10 Upon passing this course, trainees shall receive a Course Report (CF 377) and Certificate of Military Achievement (CF 289) completed IAW DAOD 5031-9 “Course Reporting and Certificates” and NPTORD 4500-51 “Course Reports and Certification”.

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

11 PLARs shall be conducted in accordance with NAVORD 4500-2 “RCN Prior Learning Assessment and Recognition”, DAOD 5031-1 “Canadian Forces Military Equivalencies Program”, and CFITES Vol 12 “Canadian Forces Military Equivalencies Programme: Prior Learning Assessment”.

CHAPTER 2 ASSESSMENT OF TRAINEE

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY THE COMMANDING OFFICER OF THE TRAINING DEVELOPMENT CENTRE

MARPAC RDIMS # 1381596
Chap, 1, approved 04 07 2023
Chap 2, 3 approved 04 07 2023

GENERAL

1 This is a pass/fail course. Trainees must successfully complete only the POs related to their occupation IAW NAVORD 11100-1 on ship represented in the following table:

Qualification	CP
Performance Objective	PO 001
Occupation	• W Eng Tech • Bosn • MAR Tech Mech • Mar Tech Elec • MAR Tech Jr.

2 Performance Checks (PCs) are based on the Performance Objectives (POs) in Chapter 3 of this QSP. Only the PCs related to the trainee's designated HV role, must be successfully completed to achieve the qualification.

3 Trainees will be developed and assessed using a combination of OJPR and summative assessment events. See Annex K for details.

4 Dress and deportment, ethics, conduct and behaviour demonstrated by the trainee shall be evaluated and assessed based on the standards laid out under Operation Honour and the LRH program. A trainee who displays inappropriate dress and deportment, ethics, and/or behaviour is considered to have demonstrated unsatisfactory conduct which may result in the convening of a Training Review Board (TRB).

PERFORMANCE CHECKS

5 Performance Checks (PCs). PCs are summative tests used to determine if the trainee has met the standard under the conditions (where possible) of the PO and hence can do the job. Supplemental PCs are administered upon failure of a PC and must be in the same format and have the same standard as the original PC.

6 PCs will be conducted after the trainees have successfully completed their OJPR package that demonstrates that they have had the opportunity to develop skills and apply knowledge related to the PO.

7 All PC failures and supplemental failures are to be handled IAW NPTORD 4500-54 Training Review Board Process. Supplemental PCs will normally be scheduled not less than 24 hours post counselling, depending on assessment resource availability

RESPONSIBILITY FOR TESTING

8 Assessment of trainees is to be done IAW CFITES Vol. 7 “Evaluation of Leaners”.

9 NFS staff provides quality control by administering and scoring all ECs and PCs IAW the standards laid out in this QSP and the assessments developed by NTDC(P).

PROGRESS MONITORING

10 Trainee progress will be monitored using formative assessment tools to help the trainee and instructional staff recognize progress or lapses in learning so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. Formative assessment is used during a phase of instruction to reinforce learning so that content is retained.

11 Monitoring learning is the responsibility of all instructional staff and instructional supervisors. Monitoring, including unsatisfactory progress, is done IAW TE SOs or SOPs and CFITES Vol 6 “Conduct of Instruction Programmes”.

12 Continuous monitoring of candidate progress is required to provide the following:

a. Early warning of trainee difficulties;

b. Feedback on the effectiveness of training;

c. Information for any required TRBs; and

d. Information to assist in completion of the CF 377 Course Report.

13 Trainees experiencing difficulty in any area of performance are to be counselled with regard to the nature of their shortcomings and suggested corrective action IAW TE SOs or SOPs. A record of trainee counselling is to be maintained by the training establishment.

14 Instructional staff shall maintain individual trainee files, including the results of all:

a. Trainee performance documentation (e.g. Weekly write-up, medical chits, dress inspection chits, conduct chits, etc.);

b. Formative assessment material (e.g. ECs, practice periods, homework, assignments and quizzes, etc.); and

c. Summative assessment material (PCs, CECs).

TRAINING REVIEW BOARD

15 TRBs shall be convened and conducted IAW with NPTORD 4500-54 “Training Review Board Process”.

16 In the event of prolonged trainee absence, TE Staff shall determine, on a case by case basis, the necessity to convene a TRB.

COURSE REPORTS

17 Course reports (CF 377) shall be prepared and distributed IAW DAOD 5031-9 “Course Reporting and Certificates” and amplified in NPTORD 4500-51 “Course Reports and Certifications”.

18 Trainees will be graded pass or fail. The narrative will indicate the trainee’s strengths and weaknesses based on PC results to provide information to the Career Manager.

19 The Course Report narrative comments could include:

a. Ability to work with others;
b. Ability to work under pressure;
c. Written and verbal communication skills;
d. Organizational skills; and
e. Leadership ability.

20 Training limitations and / or incomplete training must be included in the course report.

CHAPTER 3 PERFORMANCE OBJECTIVES AND LESSON SPECIFICATIONS

AUTHORIZATION FOR THIS CHAPTER AND/OR AMENDMENTS MUST BE GRANTED BY THE COMMANDING OFFICER OF THE TRAINING DEVELOPMENT CENTRE

MARPAC RDIMS # 1381596
Chap 1 approved 04 07 2023
Chap 2, 3 approved 04 07 2023

PO 001 – Enter High Voltage Space as a Competent Person

1. **Performance:** Enter high voltage space as a competent person.

2. **Conditions:**

   a. **Given:**

      (1) PPE;

      (2) Safety equipment;

      (3) Supervision; and

      (4) References;

   b. **Denied:** Nil;

   c. **Environment:**

      (1) Any conditions;

      (2) Day or night; and

      (3) At sea or alongside.

3. **Standard:** The high voltage safe system of work (HVSSOW) competent person shall enter an HV space as a competent person IAW refs to include:

   a. Physically reporting to the Authorized Person in Control (APC) for verification;

   b. Signing High Voltage Switching Schedule (HVSS) form IAW Annex SOP-S13B;

   c. Initialling part 1 of the PTW form IAW Annex SOP-S13C;

   d. Signing part 2 of the PTW form IAW Annex SOP-S13C;

   e. Accessing switchboard key safe;

   f. Supervising HV work performed by external contractors;

   g. Visually verifying the work area is free of hazards upon completion of work;

   h. Verbally informing the APC that the PTW is complete; and

   i. Completing completion administration.

4. References: A1, A2, A3.

5. Task Numbers: A1, A14, A15, A16.

6. Remarks:

a. Training schedule limited due to the availability of HV OJPR coaches and AOPV ship;

b. Instruction is provided online due to this PO being primarily focused as an introduction for personnel operating within HV spaces; and

c. Practical exercises are to be measured through the completion of the OJPR checklist.

EO 001.01 – Enter High Voltage Space as a Competent Person

1. **Performance:** Enter a high voltage space as a competent person.

2. **Conditions:**

   a. Given:

      (1) Computer;

      (2) DLN 3.0 access; and

      (3) References.

   b. Denied: Nil.

   c. Environment:

      (1) Any location; and

      (2) Day or night.

3. **Standard:** The trainee shall enter a high voltage space as a competent person IAW refs, by:

   a. Physically reporting to the Authorized Person in Control (APC) for verification to include:

      (1) High voltage work credentials IAW SOP-S13;
      (2) Worksite PPE is secured with PiC to include:
         (a) Hearing protection;
         (b) Eye protection;
         (c) Steel toe boots; and
         (d) Naval Combat Dress (NCD);
      (3) Personal key safe padlock is secured with PiC;
      (4) Personal key safe key(s) is secured with PiC; and
      (5) Padlock identification tag(s) is secured with PiC.

   b. Signing High Voltage Switching Schedule (HVSS) form IAW Annex SOP-S13B to accept equipment isolation after physical HV system walkthrough by APC to include:
      (1) Isolation points;
      (2) Earthing points;
      (3) HVSP devices used; and
      (4) Safety brief at POW.

   c. Initialling part 1 of the PTW form IAW Annex SOP-S13C to accept any special instructions and safety measures;

d. Signing part 2 of the PTW form IAW Annex SOP-S13C to accept equipment isolation by APC to include:

(1) Reading part 2 of the PTW form; and

(2) Visually verifying APC by placing a bare hand on all generator connections to prove the HV circuit is dead;

e. Accessing switchboard key safe to include:

(1) Placing key safe padlock on the switchboard key safe multi-lock hasp before HV work commences; and

(2) Removing key safe padlock from switchboard key multi-lock hasp upon completion of HV work;

f. Supervising HV work performed by external contractors to include:

(1) Work being completed IAW PTW; and

(2) Safety measures being followed IAW HV SSOW.

g. Visually verifying the work area is free of hazards upon completion of work to include clearing:

(1) Personnel;

(2) Tools;

(3) Instruments;

(4) Temporary earths; and

(5) Foreign objects.

h. Verbally informing the APC that the PTW is complete to include:

(1) Completing part 3 of the PTW form; and

(2) Returning parts 1 and 2 of the PTW form to the APC.

i. Performing HV power isolation during damage control event by:

(1) Opening breakers via IPMS; and/or

(2) Opening breakers with local switchboards; and

(3) Ensuring all decay times are observed before entering space.

j. Completing a limitation of access (LOA) in absence of APC.

4. **Time Required:**

a. EO – 6 x 50 min periods;

b. PC – 1 x 50 min period.

5. **Instructional Methods/Media:**

a. Online instruction (DLN 3.0);

b. Practical (OJPR).

6. **Training Aids/Resources Required:**

a. Workstation;

b. Workplace;

c. References;

d. DLN 3.0 access;

e. Computer with internet access; and

f. HMC Ship (AOPV).

7. **Test Details:**

a. This EO knowledge component shall be summatively assessed as part of EC 001.01. See Annex K Assessment Plan for details.

b. PC 001 performance shall be assessed within the OJPR package as part of Annex E. The candidate must satisfy the OJPR to pass PC 001.

8. Task Number: A1, A14, A15, A16.

9. **References:** A1, A2, A3, A4, A5, A6, A7, C1, C2

a. Remarks: Training schedule limited due to the availability of HV OJPR coaches and AOPV ship;

b. Instruction is provided online due to this PO being primarily focused as an introduction for personnel operating within HV spaces; and

c. Practical exercises are to be measured through the completion of the OJPR checklist.

10. Teaching Points:

Lesson Objective: Physically reporting to the Authorized Person in Control (APC) for verification			
Environment: Online		Time:	
Teaching Point	Method & Activities	Media & Resources	Refs
Introduction to HV work credentials IAW SOP-S13	Direct instruction	Online delivery via DLN 3.0	A1, A2, A3, A4, A5, A6, C1, C2
Introduction to ARC flash safety	Direct instruction	Online delivery via DLN 3.0	A1, A2, A3, A4, A5, A6, C1, C2
Securing worksite PPE with PiC to include: • Hearing protection; • Eye protection; • Steel toe boots; and • Naval Combat Dress (NCD);	Direct instruction	Online delivery via DLN 3.0	A1, A2, A6, C1, C2
Introduction to signing HV switching schedule (HVSS) by APC to include: • Isolation points • Earthing points • HVSP devices used • Safety brief at POW	Direct instruction, EDI via OJPR	Online delivery via DLN 3.0	A1, A2, High Voltage Switching Schedule Forms Annex A- Annex Z
Initialling part 1 of the PTW form IAW Annex SOP-S13C to accept any special instructions and safety measures	Direct instruction	Online delivery via DLN 3.0	A1, A2, PTW Form
Signing part 2 of the PTW form to accept equipment isolation by APC by: • Reading part 2 of the PTW form • Visually verifying APC with bare hand to prove HV circuit is dead.	Direct instruction	Online delivery via DLN 3.0	A1, A2, PTW Form

Accessing switchboard key safe to include: • Placing key safe padlock on the switchboard key safe multi-lock hasp before HV work commences • Removing key safe padlock from switchboard key multi-lock hasp upon completion of HV work	Direct instruction	Online delivery via DLN 3.0	A1, A2, A4, A6, C1, C2 PTW Form
Teaching Point	Method & Activities	Media & Resources	Refs
Supervising HV work performed by external contractors to ensure: • Work being completed IAW PTW • Safety measures being followed IAW HV SSOW	Direct instruction	Online delivery via DLN 3.0	A1, A2, A4, A5, A6, C1, C2 PTW Form
Visually verifying the work area is free of hazards, including the clearing of: • Personnel; • Tools; • Instruments; • Temporary earths; and • Foreign objects;	Direct instruction	Online delivery via DLN 3.0	A1, A2, A4 PTW Form
Verbally informing the APC the PTW is complete	Direct instruction	Online delivery via DLN 3.0	A1, A2, A4 PTW Form
Completing completion administration by signing as specified	Direct instruction	Online delivery via DLN 3.0	A1, A2, A4 PTW Form

Performing HV power isolation during damage control event by: • Opening breakers via IPMS; and/or • Opening breakers with local switchboards; and; • Ensuring all decay times are observed before entering space	Direct instruction	Online delivery via DLN 3.0	A1, A2, A4, A7
Completing a limitation of access (LOA) in absence of APC.	Direct instruction	Online delivery via DLN 3.0	A1, A2, A4, Limitation of Access Form

EC 001.01 - Enter High Voltage Space as a Competent Person

Assessment Activity
Strategy: Direct
Media: Online DLN 3.0
Method: Online assessment
Total Time: No prescribed limit
Environment: Any location, day or night
References: • NAVORD 11100-1 • HDW Class SEMS - SOP S13 – HV Switching Program • CSA Z462 Workplace Electrical Safety • DAOD 2007-1 - General Safety Program • Canada Labor Code Pt. 2 • General Safety Program – General Safety Standards, Volume 2 Chapter 17 - Electrical • HDW Class SSO AL 11 25 April 2020 • Standard for Electrical Safety in the Workplace 2018
Pass/Fail Criteria: 80% and trainees must achieve a SAT on all Critical checklist items (marked with an asterisk [*]). If a trainee is unsuccessful on a SAT checklist item, the assessment will cease.
Impact: • Re-test upon first unsuccessful EC. • Recourse upon 2nd consecutive unsuccessful EC
Assessment Details
Output:

Upon completion of this assessment, the trainee will have demonstrated sufficient knowledge to move onto the OJPR stage as a CP.		
Feedback / Debrief: N/A		
Assessment Resources		
OJPR coach Resources: • PC 001 OJPR Checklist • References	Learner Resources: • PC 001 OJPR Checklist • HDW Class SEMS SOP S-13 • EC 001.01 via DLN 3.0	To be Developed: EO 001.01 online courseware EC 001.01 Question bank.

EC 001.01 - Assessment Activity Checklist

Assessment Activity Checklist EC 001.01				
Candidate Name	**Assessor Name**	**Date**		**Course Session**
Item		**SAT**	**UN SAT**	**Comments**
Scenario: You are assigned to the newly commissioned HDW class vessel. You have received your onboarding package, and have been given a walk-around on the ship to familiarize yourself with the location of various spaces. You are given the role as a CP for HV. You were assigned to check oil levels, gauges, and to ensure there are no leaks present. As you are performing your rounds to check on various systems, you become aware there are a number of machinery spaces where HV equipment is located. Knowing the risks associated with HV systems, you are aware as the CP, you are required to follow certain protocols when personnel with PTW are working within these spaces. How would you ensure the safety of personnel working with PTW on HV equipment? What are the steps required to perform the function of a CP while supervision and confirming work is complete?				
1 Supervise PTW Procedure (30 mins)				
Report to APC - HV work credentials - Personal key safe padlock and keys - Padlock identification tags - Worksite PPE				

***Sign HVSS Form**
- Physical HV system walkthrough by APC to include:
  o Isolation points
  o Earthing points
  o HVSP devices used
  o Safety brief at POW				

***Sign part 2 PTW** - Read part 2 PTW form - Visually verify APC places bare hand on generator connections				

Place key safe padlock on switchboard			
*****Trainee verbally briefs assessor how to supervise external contractor to include: o Work performed IAW PTW o Safety measures followed IAW SSOW			
Assessor confirms maintenance work by external contractor is complete to trainee			
Work area is clean and clear to include: - Personnel - Tools - Instruments - Temporary earths - Foreign objects			
***Verbally inform APC work is complete** - Complete part 3 PTW form - Return parts 1 and 2 PTW to APC			
Remove key safe padlock from switchboard key safe			
Additional Comments			
Strengths:		Areas of development:	
The candidate has received a debrief and seen the content of this assessment			
Overall Assessment (circle one)		PASS	FAIL
	Name	**Date**	**Signature**
Assessor			
Candidate			

PC 001 - Enter High Voltage Space as a Competent Person Training Plan

OJPR Activity
Strategy: Individual Study, Experiential, Direct
Media: OJPR coach, real equipment, references
Method: 1 HV Training Onboarding Meeting 2 eLearning Course 3 Physical Tour of HV compartments and HV status boards on ship 4 OJPR Signature: Demonstration & Practice Activities
Total Time: 600 mins (up to, but not more than one (1) week based on HMC ship schedule and availability)
Environment: HMC Ship (AOPV)
References: • NAVORD 11100-1 • HDW Class SEMS - SOP S13 – HV Switching Program • CSA Z462 Workplace Electrical Safety • DAOD 2007-1 - General Safety Program • Canada Labour Code Pt. 2 • General Safety Program – General Safety Standards, Volume 2 Chapter 17 - Electrical • HDW Class SSO AL11 25 April 20 • Standard for Electrical Safety in the Workplace 2018
Pass/Fail Criteria: 100% completion of OJPR checklist
Impact: TRB upon unsuccessful completion of OJPR checklist within provided timeline.
Limitations / Remarks: 1. Availability of HV SMEs and AOPV ship to conduct training will impact scheduling 2. Student shall ensure they complete their OJPR package within the allocated time as scheduling permits.

3. A TRB shall be held if a student is unable to successfully complete their OJPR package within the allocated time.

Training Details

Input:

Trainee shall be posted onto an AOPV as a member of the Naval Technical Department Team, and report to SEPM.

Activity Process:

Stage 1 – HV Training Onboarding Mtg (30 mins)

- Trainees conduct a one-on-one mtg with SEPM about their HV individual training and education roadmap using HV training requirement resources and student guide.

Stage 2 – Physical Tour (90 mins)

- Trainees receive a physical tour by OJPR coach IAW with tour guide resource

Stage 3 – On Job Training

- Students shall perform all tasks specified within their OJPR package.
- Students are required to obtain SAT on ALL checklist items.

Supervising PTW Procedure – DP (90 mins) – (Demo Practice)

OJPR coach initiates the activity by providing each trainee a copy of a pre-approved HVSS, and a real PTW form that has been completed in the past. The OJPR coach then provides an overview of the PTW procedure and examples of common PTW scenarios. The OJPR coach then demonstrates while explaining how to supervise a PTW as a PiC CP.

Practice Activity

- Trainees practice completing the PTW procedure with a partner. During the activity, one partner acts as the APC and evaluates their partner using the checklist while under the supervision of the OJPR coach. OJPR coach provides ongoing feedback to individuals/groups during the activity
- Once the OJPR coach is confident trainees can independently perform the PTW procedure, the group will move onto the next step

OJPR Signature 1

- The OJPR coach will sign the trainee's OJPR package if they have successfully demonstrated proficiency in supervising a PTW procedure

Feedback / Debrief:

<table>
<tr><td colspan="3">

- Trainees conduct a training completion debrief meeting with SEPM to include:
  - Training debrief
  - Verbal confirmation trainee will be awarded the associated MITE and NQUAL for completing HV Safe System of Work course
  - Completion of certificate of appointment competent person by AE and APC endorsed SEPM

</td></tr>
<tr><th colspan="3">Training Resources</th></tr>
<tr><td>OJPR coach Resources:

• Physical Tour Guide
• PTW form
• PTW checklist
• Annex SOP S13K – Certificate of Appt Process CP
• Training schedule
• HV training student guide</td><td>Learner Resources:
• PC 001
• Safety equipment
• PTW form
• PTW checklist
• Personal Digital Device or work issued tablet
• Training schedule
• HV training student guide</td><td>To be Developed:
• CP OJPR checklist (to be within Annex E)
• Physical tour guide
• OJPR checklist to be reviewed</td></tr>
</table>

ANNEX A - MAIN REFERENCES

A = CAF or other official Canadian government publication
B = Official Allied Force or Allied government publication
C = Commercial publication
D = Other

Ref Code	Publication
A1	NAVORD 11100-1
A2	HDW Class SEMS - SOP S13 – HV Switching Program
A3	DAOD 2007-1 - General Safety Program
A4	HDW Class SSO A11
A5	Canada Labour Code Pt. 2
A6	General Safety Standards, Volume 2 Chapter 17 - Electrical
A7	Damage Control Instructions 03/20 Home Port Alongside
C1	CSA Z462 Workplace Electrical Safety 2021
C2	Standard for Electrical Safety in the Workplace 2018

ANNEX B - TASK LIST

Task No	Task Statement	Train / No Train	NT Reasoning	PO 001 - CP
A1	Supervise PTW	T		CP
A14	Issue LOA	T		CP
A15	Close LOA	T		CP
A16	Cancel LOA	T		CP
A21	Conduct HV work on dad HV equipment - Changing air filter - Changing tech water - Changing oil - Changing light bulb	NT	Prior training in corrective and preventative maintenance	CP

ANNEX C - QUALIFICATION STANDARD AND PLAN WRITING BOARD RECORD OF PROCEEDINGS

Dates: 11 Mar 2021

Location: Virtual via MS Teams

Board Members:

Position	Rank and name	Role / Unit	Dates
Chair	PO1 Aaron Reid	Sr. Electrical Propulsion Mgr on HMCS HDW	11 Mar 21
SME	PO1 Marvin Murphy	LOTO Coordinator, FMF Cape Scott	11 Mar 21
SME	PO2 Michael Lamanque	Sr. Mech Specialist on HMCS HDW	11 Mar 21
TDO	SLt Sebastian Seo	TDO, Training Dev Centre Pacific	11 Mar 21
Secretary	Ms. Marilyn Carey	Civilian	11 Mar 21

Board Objective: To finalize the HV Safe System of Work QSP

Decision/Event	Discussion / Detail
Day 1: March 11, 2021	
SMEs	Given high voltage is so new to the RCN, there are no real SMEs for HV in the RCN
No task list	No approved HV task list from AOPV PMO or Occupational Authority or updated MAR Tech JTAR (last updated in 2019). Board referenced SEMS SOP-S13 document to identify HV tasks and based on reality of the job.
Exclude devising sketches for HVSS in QSP	PO 2 Lamanque commented that there may be some circumstances where an APC may need to devise a hand written sketch outside of the sketches provided on the pre-authorized HVSS form. Discussion about need to include this in QSP. HVSS pre-approved forms to be verified through QA on Mary Brooke (Seo). Decision: No train due to infrequency of event, prior training

Decision/Event	Discussion / Detail
Exclude HV maintenance tasks on QSP.	Discussed whether performing non-HV maintenance tasks should be inserted into QSP PO 002 (AP). Reid: methods for maintenance not changing depending on HV equipment. Only change is locking out. Lamanque: Maintenance is done HV Dead. Seo: MarTechs have prior training on maintenance. Murphy: I concur.
	Lamanque: Assume HV Live state until proven dead. Should this be included? Reid: Yes, I agree
Exclude performing SFT insulation testing	For insulation testing, prior learning for MAR Tech elecs on smaller circuits – already know how to do this (Reid). In reality, APs have been supervising contractors – this is also what the contract stipulates with external contractors (Murphy). Exclude from QSP
Exclude performing maintenance during SFT.	PO 002 (6) Discussion – SFT while at Sea. Lamanque: May be a situation at sea where ship staff will conduct work under an SFT. Reid: Yes, SFT includes running the machinery for testing purposes, e.g. propulsion converters updates on recent sail. Seo: Did you get formal training through Calhoon? Reid: Everything I've learned has been on my own. That's what we do as electricians.
Remove "HV safe" from PO 002 d.(6)	Reid: Only APC is to make the informative pipe for "HV Safe" during a damage control event, but the caveat is Duty Engineer would need to make this pipe while alongside and the ship is HV Live. Murphy: This is only related to when the ship is HV Live.
Change "switchboard safe" to "switchboard register" PO 002 d.(7)	Decision to change to "switchboard register" instead of "switchboard safe" because there is no place to mount the document safe on the wall on the AOPV ship because of fuel tanks, etc. (Reid)
PO 002 e. (10)	Discussion of role of safety member. Part of electrical training to never work on live circuits alone. If you need to test something, you would have safety member with 10 ft safety cane in case of accident (Murphy). Action of safety member = NO TRAIN (Seo). "Leave this one with me" (Seo)
PO 002 e. (8)	Placing all HVSS keys in switchboard key safe moved from (10) to (8). Important to put in QSP (Reid).
PO 002 f. (a.iv)	Added "visually inspecting worksite PPE." Worksite PPE, not necessarily Arc Flash rated PPE needed. (Reid)
PO 002 f. (4.c.)	Discussion of signing off on both copies of PTW, including PiC's copy. (Murphy) Both need to be completed. Added f. (4.c.) that one completed copy of PTW needs to go into switchboard register.
PO 002 g. (2)	AP is always the PiC but they can either undertake or supervise – focus on supervising (Seo). Our standard for this QSP is supervising, not doing. (Seo) Confirmed by Reid.

Decision/Event	Discussion / Detail
PO 002 h. 2(b) Are you always cordoning off the area with caution tape? (Murphy)	Not all the time, spaces are so big – never did an LOA in MCR, all LOAs so far main machinery room. These spaces are massive, HV is in one small area. Only once cordon off (Reid). There should never be concurrent work going on (Murphy). Decision to train personnel to cordon off. (Seo) Confirmed by Reid.
PO 002 i.	Added "Update HV Stateboards" (Murphy)
PO 002 k.	Do we need to train ppl: Managing all HV equipment, devices and accessories as an APC to include: (1) Visually inspecting all HVSP equipment, devices and accessories periodically IAW Equipment Inspection Guidelines to include: (a) Cleanliness; (b) Damage; and (c) Security. (2) Tracking HVSP equipment, devices and accessories accessed by personnel; and (3) Ordering replacement HVSP equipment, devices and accessories. Decision: No. Already trained duty engineers. This is no different. (Reid, Lamanque) NO TRAIN – Prior learning
PO 002 l.	Conducting a SEPM handover IAW Annex SOP-S13G with the oncoming Senior Electrical Propulsion Manager No Train as SEPM handover is same as APC Handover. It would be AE level. Completing HV safety program dev and evaluation technique profile – task rarely performed?
PO 002 m.	Completing HV safety program development and evaluation technique profile form IAW Annex SOP-S13O Not sure if this needs to be trained. Personally, I will do it. (Reid). Seems to be an experiential learning piece (Lamanque) NO TRAIN – task is done rarely. No SME experience with this. (Seo)
PO 001-3a.	Remove "Visually inspect and don PPE" from CP training. I've never seen a situation where the CP needs to don the PPE (Reid).
PO 001-3	Remove printing HV forms from CP level. AP would print these forms.
PO 001-3	"Completing" HVSS changed to "Reviewing" HVSS (Murphy).
PO 001-3	Conducting HV work on dead HV equipment to include: (a) Changing air filter; (b) Changing technical water;

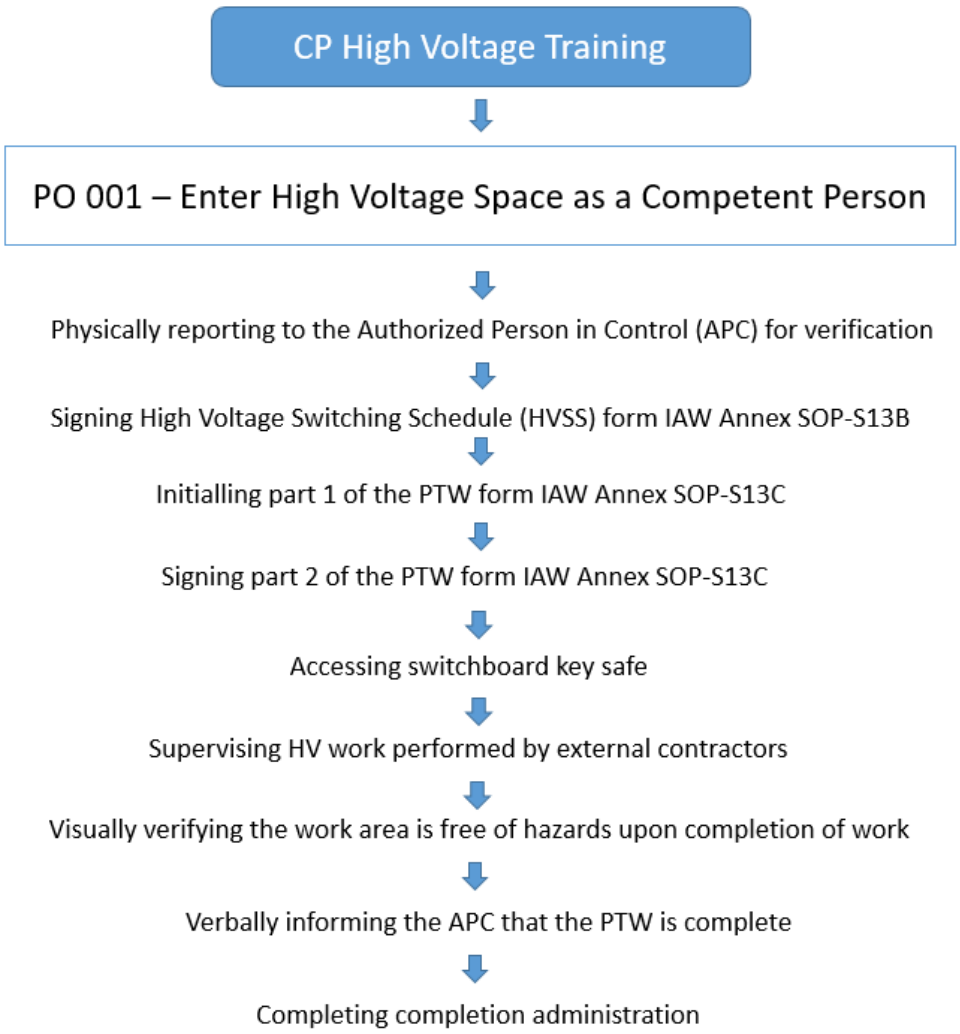
Decision/Event	Discussion / Detail
	(c) Changing oil; and (d) Changing light bulb No train since MarTech Mechs already have prior training on CM and PM. (Reid, Lamanque, Murphy)
PO 001-3	Remove "Supervising work performed by HV aware personnel to include: a. Work being completed IAW LOA; b. Safety measures being followed IAW HV SSOW; and c. Work area is clean upon completion of work.
PO 003-3a,b	Train (a) based on SME response. AE must know the process. (Reid, Murphy) (b) These are different processes – all separate certificates, handwritten (Reid). CP Certification: WEng CANNOT work on any HV equipment; Mec's can only work on DEAD HV equipment; AP's can work on LIVE HV equipment– long process, must train (Reid).
PO 003-3c,d	Authorizing all High Voltage Switching Schedule (HVSS) work by: Visually checking the HVSS form is complete; Initialing step 1 of HVSS form. Authorizing all HVSP work by: Visually checking HVSP forms are complete; Signing HVSP forms to include: (a) PTW; (b) SFT; and (c) HVSS No Train (c) – leaning on APC, AE needs to know what work is being done and rationale behind it (Reid). 90% of work, they know it's coming (Murphy). Already trained in AP module (Seo). No Train (d) – Already trained in AP course. (Seo) Confirmed by Reid, Lamanque and Murphy.
PO 003-3e,f	Visually inspecting HVSP PPE and testing equipment maintenance records periodically; Conducting monthly review of switchboard register to include: (1) Visually inspecting volumes 1 and 2; (2) Signing in volume 1 and 2 binders; and (3) Forwarding volumes 1 and 2 to CO for quarterly review and signature. No Train (e) – just making sure records are being kept (Reid, Lamanque) Records not signed off – nothing to train, but need to be aware (Murphy). These should fall under "Responsibilities" for AE training (Lamanque, Seo)

Decision/Event	Discussion / Detail
	No Train (f) – easy task (Seo), knowledge base (Reid). Navord indicates CO signs monthly. (Murphy)
PO 003-3g, h	Conducting periodic on-site safety audits by: (1) Visiting an HV work site; (2) Visually inspecting safety measures are being followed. Responding to HV near miss incidents to include: (1) Completing HV Near Miss Report SOP-S13P; (2) Handing-over report to Unit General Safety Officer; (3) Placing copy in switchboard register binder. NO Train (g) – prior learning in AP (Lamanque) NO Train (h) – admin (Lamanque), UGSO generates report (Murphy) simple task (Seo) AE Half day course, knowledge based on Sultan with test at the end (Reid)
PO 003-3 i,j	Obtaining permission from CO for the emergency removal of HVSP locking devices. Authorizing SEPM and APC handover certificate to include: a. Visually checking handover certificate forms to include: b. Senior Electrical Propulsion Manager handover certificate SOP-S13G c. APC handover certificate SOP-S13H Signing part 3 certificate of transfer No Train (i) simple task and/or not done rarely. Covered in LOTO system (Reid) No Train (j) SME's agreed.
PO 003-3 k,l	Approving the master key register by: a. Visually checking SOP-S13M form is complete; and b. Initialing the SOP-S13M form. c. Verbally briefing the captain on ship of deviations to approved safety rules. No train (k) – simple task, rarely done. Keys will never change b/c we use same keys and locks all the time. Takes awhile to set up. HDW every key and lock has one place to go. No train (l) – Already have training on how to talk to CO (Reid, Lamanque, Murphy)
AE Assessment?	Test should ask for references and where to find them (Murphy). Reid agreed. Scenarios? No need to request HMS Sultan Test – not relevant to AOPV (Reid)

Decision/Event	Discussion / Detail
Knowledge of references?	SEMS S-13, CSA bits and pieces, Canada Labour Code bits and pieces, Discuss later to figure out which pieces.
Day 2: March 17, 2021	
Assessment of PO 002	Regarding assessment, should the AP trainee be assessed on: (a) When closing the PTW, visually inspect work area, including clean work area, all tools and materials removed? (b) When closing the SFT, visually inspect work area, including clean work area, all tools and materials removed? (c) When closing the SFT, inform the AE and Duty Tech that work is complete, and DG 1 breaker is now available? Yes to a, b, c. These are all important duties for the APC. (Murphy, phone call with Marilyn). After the completion of a PTW or SFT, the PiC is responsible to ensure the worksite is cleaned up and the APC will inspect the worksite to ensure that it is complete before all final signatures. (Lamanque, email)

ANNEX D - QSPWB SCALAR

The mapping of the task analysis from the QSPWB is detailed in the standard paragraphs of the PO in Chapter 3 of this QSP.



ANNEX E - ON JOB PERFORMANCE RECORD (OJPR)

On-the-Job Training (OJT)

OJT is a component of formal training in which personnel learn tasks while performing their actual duties on a job. It is guided by performance objectives and checks.

The On-the-Job Performance Record (OJPR)

As part of HV interim training, the designated AOPV ship is responsible for conducting OJT, assessing learners and documenting whether the objectives were attained in the learner's OJPR. The designated AOPV ship reports the results of the OJT back to the training establishment responsible for the training program. The training establishment enters the results on the individuals' record and awards the qualification upon successful completion.

Prior to commencing OJT, personnel are expected to have completed stages 1 to 3 below:

Stage 1 – HV Aware eLearning Brief
Stage 2 – CP eLearning Course
Stage 3 – HV Physical Tour on ship
Stage 4 – OJT on designated AOPV

High Voltage - Competent Person Qualification

Name and Initials	Rank	SN

Date Training Started	Expected Date of Completion

THE OJPR MUST BE COMPLETED IN ACCORDANCE WITH THE INSTRUCTIONS BELOW. INCORRECT OR INCOMPLETE OJPR WILL BE RETURNED FOR RE-SUBMISSION.

INSTRUCTIONS

One copy of this OJPR, hereinafter referred to as the master copy, shall be made up for each individual beginning OJT. It shall be kept and maintained by the supervisor. The Training Establishment shall make a duplicate copy for retention by the trainee for the purpose of recording and reviewing individual progress. The TE for the OJPR must maintain Training Tracker fields as outlined in Monitor Mass.

As the trainee completes each Performance Objective (PO) to the required standard, the Training PO or Officer will date and sign the applicable item on the master and duplicate copies of the OJPR sheet and inform the individual of the action taken.

It may not be possible to conduct OJT for some POs if, for example, the necessary equipment is not fitted on a ship, is not on issue to the unit, or is otherwise unavailable. If the training PO or Officer encounters any of these issues they are directed to contact Standards at the appropriate TE for guidance.

If an individual demonstrates the ability to perform additional duties not specified in this QSP, an appropriate entry shall be made under “Additional Duties” at the end of the OJPR.

PROGESS REVIEW

Progress review shall be carried out quarterly by the Divisional Officer and Section Head (NCMs) or Training Officer (Officers)

PC 001 Competent Person

ID	Practice Activities	OJPR Reviewed By (Rank, Position, Name)	Signature	Date
1	Tour of HV Spaces			
2	Able to recite and point out HV/potential HV spaces on ship's drawing			
3	Able to recite and point out all HV equipment			
4	recite actions taken for an DC event in an HV space			
5	operate HV breakers from IPMS/HV SWBD			
6	Performing LOA procedure as an DE			

TRAINING COMPLETION CERTIFICATE

Date training completed:

CERTIFICATION OF QUALIFICATION

I certify that _____ has completed training to the standards stated in the HV Safe System of Work QSP: PO 001 – Enter High Voltage Space as a Competent Person. The member is qualified to carry out the duties of an Competent Person.

OJPR Coach Signature: _____ Date: _____

Naval Technical Officer Signature: _____ Date: _____

Training Activity Details

Stage 5 – OJT

<u>Activity 1: Tour of HV/Potential HV spaces</u>

Demonstrate

- A member of the Naval Technical Department (NTD) with the qualification of CP or higher is to give an in-depth tour of all High Voltage and Potential High Voltage spaces. They must show all HV equipment and HV cables. This includes potential HV spaces in the event of a DC event.

Practice Activity

- When arriving at AOPV, it is important for a tour to be given at the earliest availability.

OJPR Signature 1

- The OJPR coach will sign the trainee's OJPR package if they have successfully completed a tour of all HV spaces.
-

<u>Activity 2: Reciting all HV/potential HV spaces</u>

Demonstrate

- Trainees should be able to recite and show location on ship's drawing where all HV/Potential HV spaces are.

Practice Activity

- If the ship is "HV Live" trainees should ask for availability of NTD personnel to give more tours to better understand HV spaces. If the ship is "HV Dead" then the trainees are to do self guided tours to better understand the layout of HV spaces and equipment..

OJPR Signature 2

- The OJPR coach signs the trainees' OJPR package if they have successfully demonstrated proficiency in reciting and pointing out on ship's drawing.
-

<u>Activity 3: Listing and recognizing all HV equipment</u>

Demonstrate

- Trainee's must be able to identify and recite all HV equipment.

Practice Activity

- Trainees can do self-guided tours when the ship is “HV dead.” If the ship is “HV Live” then trainee’s are to company CP qualified members of the NTD to get better acquainted with the HV equipment.

OJPR Signature 3

The OJPR coach will sign the trainee’s OJPR package if they have successfully demonstrated the identity and location of all HV equipment.

<u>Activity 4: Responses to HV Spaces during an DC event</u>

Demonstrate

- Being able to recite and know the exact steps for responding to a DC event in an HV space when the ship is “HV Live”.

- Being able to recite the “Decay Times” for each space and for each HV equipment.

Practice Activity

- Trainee’s is to familiarize themselves with the “Decay Times” of Spaces and their equipment.

OJPR Signature 4

The OJPR coach will sign the trainee’s OJPR package if they have successfully recited all the “Decay Times” for the spaces and all its HV equipment.

<u>Activity 5: Operating HV Breakers remotely and locally</u>

Demonstrate

- Trainees shall operate HV Breaker remotely via IPMS or OBITS. The trainees shall be able to explain how Power Management System (PMS) works.

- The trainee with the supervision of an AP qualified member is to demonstrate how to operate an HV breaker locally.

Practice Activity

- Trainees shall be given the opportunity to practice operating HV breakers when the time presents itself. When possible, the IPMS Tech would set up an OBITS trainer for IPMS so that the trainee can practice synthetic operation of HV breakers.

- An AP qualified personnel shall explain and go over the local operation of the HV breaker. If the opportunity presents itself, the trainees shall practice local operation of the HV breakers with the permission of Command and supervision of an AP qualified person.

OJPR Signature 5

The OJPR coach will sign the trainee's OJPR package if they have successfully demonstrated proficiency in operation of HV breakers remotely and locally.

Activity 6: Performing Performing LOA Procedure as an "Duty Engineer"

Demonstrate

- OJPR coach initiates this activity by providing each trainee with a completed LOA example. The OJPR coach demonstrates while explaining how to perform a LOA procedure.

Practice Activity

- OJPR coach provides trainees with a blank LOA form and new LOA scenario. Trainees complete the LOA procedure individually while under OJPR coach supervision. Trainees then complete the form and submit to OJPR coach for evaluation. The OJPR coach evaluates the form and provides individual feedback to each trainee

- Once the OJPR coach is confident trainees can independently perform the LOA procedure, the group will move onto the next step.

OJPR Signature 6

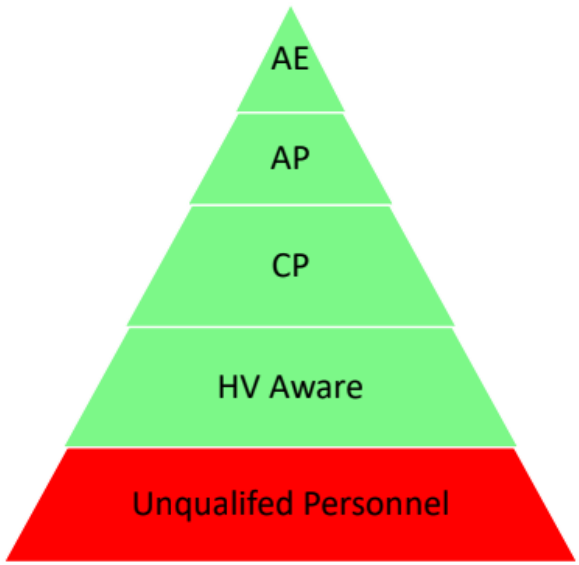
- The OJPR coach will sign the trainee's OJPR package if they have successfully demonstrated proficiency in performing the LOA procedure as an DE.

ANNEX F - ACRONYMS AND ABBREVIATIONS

Acronym / Abbreviation	Term or Definition
AE	Authorized Engineer
AP	Authorized Person
APC	Authorized Person in Control
CME	Circuit Main Earth
CP	Competent Person
HV	High Voltage
HVSP	High Voltage Switching Program
HVSS	High Voltage Switching Schedule
LOA	Limitation of Access
NTD	Naval Technical Department
PiC	Person in Charge
POW	Point of Work
PPE	Personal Protective Equipment
PTW	Permit to Work
SEMS	Safety and Environmental Management System
SFT	Sanction for Test
SSO	Ship’s Standing Orders
SSOW	Safe System of Work

5. HV Qualifications and Training Requirements

5.1 Qualification Levels



Authorizing Engineer (AE)
- Naval Technical Officer
- Senior Electrical Propulsion Manager

Authorized Person (AP)
- Leading Seaman and above Marine Technicians - Electrician

Competent Person (CP)
- All members of the Naval Technical Department, and other personnel where appropriate may also hold this qualification

HV Aware
- Entire ship's company, DND employees, regular contractors, and routine visitors

Unqualified Personnel
- All other visitors

Figure 5.1 – HV Qualification Hierarchy

MARPAC RDIMS # 1381596
Chap 1 approved 04 07 2023
Chap 2, 3 approved 04 07 2023

		Distributed Learning	OJPR	
Competent Person **1 day**	Stage 1 (30 mins)	Stage 2 (300 mins)	Stage 3 (90 mins)	Stage 4 (600 min – up to no more than five (5) days)
	Posted to AOPV Ship	PO 001 Perform HV SSOW Procedures as a Competent Person	AOPV HV Orientation	OJPR Signature: Demonstration and Practice Activity
	• HV Training Onboarding Mtg • HV Training Requirements & Schedule • HV Training Student Guide	Lesson Objectives TBD for serial 2 as HDW ship conducted F2F CP training as part of serial 1	Physical tour of HV compartments and HV status boards	Supervising PTW procedure(s)

ANNEX H - ADDITIONAL TRAINING RESOURCES

In accordance with Chapter 1 para3-4, add any extra tables or paragraphs used for training resources in this annex. (Annex H not to be deleted)

ANNEX I - COSTING INFORMATION

Include information about cost in whichever format is applicable (Not to be deleted).

Estimates only, based on CAF Cost Factors Manual FY 2012/13
verify that this link is the most updated version

**5 year total based on one course per year INCLUDING Initial Set-Up. Costs may be adjusted accordingly.

ANNEX J - COURSE HISTORY

This annex shall NOT be deleted. It must be annotated with every update or modification to this QSP. This is in addition to making the appropriate entry in the Record of Amendments table. Details on course history issues such as explanations of amendments, pertinent findings, trends, resource problems and modifications shall be captured in the table below. Focus should be given to both excellent and questionable issues of the course. Capturing important course history information is a useful managerial tool for amendment considerations by future course developers.

Reason for amendment (with date)	Details on course history
QSPWB (Dates)	(Explanations of amendments, pertinent findings, trends, resource problems and modifications)

ANNEX K - ASSESSMENT PLAN

OJPR/ EC	Scope Covered	Purpose	How	Pass / Fail Criteria	Impact
OJPR	PO 001	Demonstrate proficiency in supervising PTW procedure	OJPR signature	SAT on ALL OJPR items	TRB upon unsuccessful completion within provided timeframe
EC 001.01	PO 001	To assess the trainees' knowledge before moving to OJPR package as a CP.	Online assessment	80%	Re-test/Recourse upon 2nd unsuccessful EC

ANNEX L - TEL PLAN

PO	COA 1 TEL Delivery	COA 2 Interim Delivery	Impact if TEL not implemented - Realized benefits - E/W Coast Delta
PO 001	CP eLearning Course	OJPR on ship	Impact on resources (availability of SMEs) to deliver F2F interim training

MARPAC RDIMS # 1381596
Chap 1 approved 04 07 2023
Chap 2, 3 approved 04 07 2023

ANNEX M - TARGET POPULATION AND TRAINING ESTABLISHMENT ANALYSIS

The following information was compiled during the QSPWB (date) to give context for developers. It can be removed from the QSP if desired after initial course development.

Key Target Population Details - (Course name) candidates

“The learners' level of skill and knowledge with respect to the performance objectives (POs) is a factor which depends mainly on previous related training or education, and experience. It may also be affected by strong opinions or attitudes about the subject matter. Information should be obtained on the learners” -CFITES Vol 4

Characteristic	Target population details
General data	
Potential group size (based on new AOPV ship): • CP: 15 Expected annual throughput: 25 based on untrained full AOPV NTD crew Location(s) of the intended learners: Currently in HFX for first two AOPV ships and then ESQ once third AOPV ship arrives west coast Access to computers (job): access to NFS computers and tablets Gender: Male & Female Age: 19 to early 50s Trades: MSEO, MAR Tech Elec, MAR Tech Mech, WEng Tech	
Learning skills and preferences	
Competencies (general)	CP: Mix of trained technicians to new members
Reading and writing level	CP: Strong
Preferences, attitudes towards specific type of learning material and equipment (i.e. TEL, aversion to lectures)	Open and flexible in using TEL

Characteristic	Target population details
Educational level	University, College, High School
Previous military training experience	CP: RQ-S3 to RQ-CPO2 (1 to 20+ yrs)
Current subject-matter competence	
Experience with target performance	CP: None
Ability to perform any part of the POs	CP: Very High
Perception of the impact of achieving the POs on self, work or career	Excellent
Job/career stage	CP: early to late career (1 to 20+ yrs)
Social Characteristics	
Relationship to peers	Excellent
Tendencies toward cooperation or competition	Cooperation